

EAM310 · EAM DIAGNOSIS I · VUIM · SUMMER 2026

Study Guide

Complete Pulse Diagnosis Reference — Week 3

Prof. Daniel Schrier, D.O.M., L.Ac.

Week 3 · Pulse Diagnosis · Summer 2026

Jonathan Centeno · D.AcHM Candidate · VUIM 2026–2030

What this is

The complete Week 3 pulse reference — everything you learn *from*, in one document. Narrative first (why the system works), then the full reference (all 28 qualities, all 127 combinations), then method and technique, then the discriminations and safety findings that carry the most weight.

The seven things that matter most

If you retain nothing else: **1.** The six Nan Jing positions. **2.** The three depths, and how to find the middle one. **3.** Floating is *not* always Exterior. **4.** Huǎn at 4 beats/breath is *healthy*, not Slow. **5.** Knotted vs. Hasty vs. Intermittent. **6.** The pulse confirms — it never stands alone; ask questions. **7.** Floating, Deep, Slow, Rapid, Full are the five to own with your hands first.

PART I · Narrative — Why Pulse Works This Way

1 Context — Why Pulse Comes After the Eight Principles

Pulse diagnosis is the third of the Four Pillars you're building, and the first one that gives you a second, independent channel of evidence for the categories the Eight Principles already taught you: Exterior/Interior, Cold/Heat, Deficiency/Excess, Yin/Yang. When pulse and the rest of the picture agree, your diagnosis is confirmed. When they disagree — a patient with cold hands who is actually running a True Heat pattern underneath — that disagreement is often the single most useful piece of information you get all session.

Historical Background

Nan Jing 難經 (Classic of Difficulties) established taking the pulse at the radial artery across three levels — superficial, middle, deep. *Su Wen* 素問 originally described nine regions: three on the head, three on the hands, three on the feet. Wang Shuhe 王叔和, a physician in the imperial medical service, composed the ten-volume *Pulse Classic*, describing 24 pulse qualities and positions with Zang-Fu resonance and diagnostic significance — the direct ancestor of the 28-quality system taught today.

Quiz 3 High-Yield Targets

Six positions (L/R × Cùn/Guān/Chǐ) · Five finger movements · Normal pulse (胃神根) · Seasonal pulses (all four) · Floating/Deep, Slow/Rapid, Empty/Full combinatorial logic · True Heat/False Cold vs. True Cold/False Heat

2 The Six Positions — 寸關尺 Cùn Guān Chǐ

Radial pulse diagnosis as standardized today comes down through the *Nanjing* (Classic of Difficulties), giving each wrist three positions along the twelve regular channels. Schrier's slides use the full Nan Jing mapping

below — note that Kidney is **not** split into Yin/Yang across the two Chǐ positions here; instead, Chǐ pairs Kidney with its Nan Jing channel partner on each side.

Position	Left	Right
寸 Cùn (1st, Front)	Hand Taiyang / SI · Hand Shaoyin / HT	Hand Yangming / LI · Hand Taiyin / LU
關 Guān (2nd, Middle)	Foot Shaoyang / GB · Foot Jueyin / LR	Foot Yangming / ST · Foot Taiyin / SP
尺 Chǐ (3rd, Rear)	Foot Taiyang / BL · Foot Shaoyin / KD	Hand Shaoyang / TB · Hand Jueyin / PC

Quick Recall — Left Cùn=HT(SI) · Guān=LR(GB) · Chǐ=KD(BL)	Quick Recall — Right Cùn=LU(LI) · Guān=SP(ST) · Chǐ=PC(TB)	Burner Logic Front=Upper Burner · Middle=Middle Burner · Rear=Lower Burner
Reading Order Whole → Spirit/Stomach-Qi/Root → Levels → Strength → Quality → Position		

Heads Up: Different Schools, Different Maps

Not every lineage maps the Yang (hollow) organs the same way onto the six positions, and some modern clinical shorthand simplifies the two Chǐ positions to "Kidney-Yin / Kidney-Yang" rather than using the full Nan Jing organ pairings above. Know the Nan Jing mapping first, since it's the version taught in lecture — treat the simplified Yin/Yang shorthand as a different (and looser) convention, not an error if you see it elsewhere.

3 Method — Five Finger Movements

FCM Ch.25 names five distinct finger movements. Students often only recall four — know all five:

#	Movement	Checks for
1	Lifting	Superficial level — light touch
2	Pressing	Deep level — firm touch to the bone
3	Searching	Rate
4	Pushing	Shape
5	Rolling	Length

Rate is traditionally counted against the *practitioner's* own breath, not the patient's — the assumption being your breathing is the more stable reference point.

4 The Normal Pulse — 胃神根

胃 Wèi — Stomach-Qi

Flowing, gentle, harmonious movement

神 Shén — Spirit

Strong, rhythmical — vitality present in the beat

根 Gēn — Root

Depth felt at Chǐ — Kidney Qi intact under firm pressure

Classic Image

A cow chewing its cud — soft and firm at once, like a cat's underbelly

5 Seasonal Pulses

Each season carries a pulse quality that is normal *for that season* in a healthy person. The recurring trap question: "is this pulse abnormal?" — the answer depends entirely on whether other signs and symptoms accompany it.

Season	Pulse	Dates	Why
Spring	Wiry 弦	Feb 4 – May 4	Yang Qi wants to rise; Cold-Yin nature still obstructs release — shouldn't be <i>too</i> wiry
Summer	Flooding 洪	May 5 – Aug 6	Yang Qi uninhibited, spreading freely, at its peak
Autumn	Hair-like/Floating 毛	Aug 7 – Nov 6	Yang Qi desiring to be restrained and collected — "dainty, like elm pods falling"
Winter	Stone/Deep 石	Nov 7 – Feb 3	Yang Qi gathered and stored; sinks like a stone

Clinical Caution

A Flooding pulse in winter is a genuine warning sign, not a seasonal variant — Yang should not be surging in winter. Schrier's teaching case: a patient with a Flooding pulse in winter declined treatment; by summer, the already-excessive Yang compounded with the season's natural Yang rise and the patient suffered a cerebral hemorrhage. Rule: a seasonally "expected" pulse is only truly seasonal when no other signs or symptoms point to pathology.

6 The Combinatorial Logic — Building Patterns Like Lego

Most of the 28 pulse qualities are not learned as 28 isolated facts. Three foundational pairs — Floating/Deep (location), Slow/Rapid (speed), Empty/Full (strength) — combine with secondary qualities to generate nearly everything else.

Floating vs. Deep — Location

Combination	Meaning
Floating + Tight	Exterior Wind-Cold
Floating + Rapid	Exterior Wind-Heat
Floating + Empty	Yin Deficiency (not Exterior — see trap below)
Deep + Slow	Interior Cold
Deep + Rapid	Interior Heat
Deep + Weak	Yang Deficiency
Deep + Wiry	Blood stasis, or a Chong-vessel issue

Watch For: "Floating = Always Exterior"

False. Floating also appears in Yin Deficiency: Yin fails to anchor Yang, which floats to the surface — presenting Floating, Thin, and Rapid, with night sweats, five-palm heat, malar flush, dry throat worse at night. Differentiator: true Exterior floating comes with simultaneous aversion to cold and fever. The moment that pairing disappears, the pathogen has moved Interior — and Floating should disappear with it.

Slow vs. Rapid — Speed

Combination	Meaning
Slow + Weak	Empty Cold (Yang Deficiency)
Slow + Full	Full/Excess Cold — chronic painful conditions
Slow + Choppy	Blood Deficiency and interior Cold
Rapid + Full	Full Heat
Rapid + Empty/Floating	Empty Heat (Yin Deficiency)
Rapid + Wiry	Liver Fire — "the string about to snap"
Rapid + Slippery	Phlegm-Heat or Damp-Heat

What to Actually Remember — Sections 5–6

- Spring=Wiry · Summer=Flooding · Autumn=Hair-like · Winter=Stone/Deep.
- Floating/Deep=location, Slow/Rapid=speed, Empty/Full=strength — the three pairs that generate most combinations.
- Floating is not automatically Exterior — check for simultaneous aversion-to-cold-and-fever first.

7 Individual Pulse Qualities — Sensory Anchors

These qualities round out the 28. Exam questions often quote the classical sensory image rather than the clinical meaning — know both.

Quality	Sensory image	Core meaning
Slippery 滑	A pearl rolling on a plate	Phlegm, Dampness, food accumulation — or pregnancy
Choppy 澀	A knife scraping bamboo	Blood Deficiency or Blood Stasis; fluid exhaustion
Wiry 弦	A taut bowstring	Liver-Qi Stagnation, pain, Phlegm; the normal Spring pulse
Thin 細	Angel-hair pasta vs. thick udon	Blood Deficiency
Minute 微	A thread that comes and goes	Severe Qi/Blood depletion — more extreme than Thin
Soggy 濡	Wet, soft, floating, without strength	Dampness with underlying Deficiency

Clinical Judgment Call — Slippery and Pregnancy

A pronounced Slippery pulse in a woman of reproductive age is a classic (not definitive) sign of pregnancy, strengthening as pregnancy advances. It can just as easily indicate food retention or Phlegm-Damp generally. Ask indirectly — "are you trying to conceive?" rather than "are you pregnant?"

8 True/False Heat and Cold — Where Pulse Breaks the Tie

This is the material most likely to appear as a clinical vignette: a superficial presentation that contradicts the underlying reality.

False Heat, True Cold

Malar flush only, wants to cover up, warm-drink preference, cold limbs, pale wet tongue. Pulse: Floating-Empty. Interior is genuinely Cold.

False Cold, True Heat

Cold limbs despite genuine interior Heat. Thirst, irritability, constipation, dark urine, red tongue/yellow coat. Pulse: Deep-Forceful.

Tie-breaker: tongue moisture, thirst pattern, urine color — regardless of what the limbs are doing.

What to Actually Remember — Sections 7–8

- Slippery=pearl on a plate → Phlegm/Damp/food, or pregnancy. Choppy=knife on bamboo → Blood Deficiency/Stasis. Wiry=bowstring → Liver-Qi Stagnation.
- False Heat/True Cold: Floating-Empty pulse. False Cold/True Heat: Deep-Forceful pulse.
- Tongue moisture + thirst + urine color settle the tie whenever the limbs mislead.

9 Full Reference — All 28 Pulse Qualities

Sections 6–8 walked through the highest-yield qualities with their combinatorial logic. This is the complete, descriptive set of 28 — sensation, classical image, and full indications for each — in Schrier's own lecture order and language, sourced directly from his slide deck rather than a generic textbook list.

Quality	Sensation & Classical Image	Indications
1. Floating 浮 Fú	Felt distinctly at the superficial depth; force decreases slightly deeper but never feels empty. <i>Classical image:</i> Wood floating on the surface of water; a subtle breeze across a bird's back; scallion leaves rolled lightly between the fingers.	Invasion of external Wind. Yin Deficiency (an interior condition presenting this way).
2. Deep/Submerged 沉 Chén	Located deep next to the bone; barely felt with light or moderate pressure; only felt pressing to the bone and tendons. <i>Classical image:</i> A stone thrown into a well, sinking to the bottom.	Pathogenic factor in the Interior (Deep-Full). Yang Deficiency (Deep-Weak).
3. Slow 遲 Chí	Less than three beats per breath, about 40–60bpm, regular rhythm.	Cold pattern. Empty-Cold (Slow+Weak). Full-Cold (Slow+Full).
4. Rapid 數 Shuò	Over five beats per breath, 90–140bpm, regular rhythm.	Heat pattern. Full-Heat (Rapid+Full). Empty-Heat (Rapid+Floating-Empty).
5. Empty/Deficient 虛 Xū	Wide but without strength; forceless at all three positions under light or heavy pressure; a sense of emptiness.	Qi Deficiency.
6. Full/Excessive 實 Shí	Felt at all three positions and all three depths; long, wide, and forceful.	Full pattern. Full-Heat (Full+Rapid). Full-Cold (Full+Slow).
7. Slippery 滑 Huá	Feels quick though not rapid; slippery and smooth; rounded inside the vessel, as if oily. <i>Classical image:</i> A pearl rolling on a plate.	Phlegm. Dampness. Retention of food. Pregnancy.
8. Choppy 澀 Sè	Seems slow and is not smooth; vessel is thin; rhythm irregular in force and fullness; feels rough, jagged. <i>Classical image:</i> Scraping bamboo with a small knife.	Blood Deficiency. Exhaustion of Body Fluids.
9. Long 長 Cháng	Smooth and straight; felt extending distally and proximally beyond the usual positions.	Heat patterns.
10. Short 短 Duǎn	Cannot be felt in all three positions; absent at the distal and/or proximal position.	Severe Qi Deficiency. Stomach-Qi Deficiency.

11. Overflowing/ Flooding 洪 Hóng	Wide and forceful, though it fades gently; vessel wide and thick. <i>Classical image:</i> Ocean waves that come strongly but leave calmly.	Full-Heat. Empty-Heat (Overflowing-Empty).
12. Fine/Thin 細 Xì	Very thin, like a fine thread, but distinct and clear without interruption of rhythm under pressure.	Blood Deficiency. Dampness with severe Qi Deficiency.
13. Minute/Faint 微 Wēi	Extremely thin and forceless; seems to be felt, then not felt; pressing deeply the vessel becomes vague.	Severe deficiency of Qi and Blood.
14. Tight 緊 Jǐn	Rises and falls with strength, vibrating left and right; feels twisted like a thick rope. <i>Classical image:</i> Fingers on a highly stretched twisted cord, snapping back and forth.	Cold — Exterior (Tight-Floating), Interior Full-Cold (Tight-Full-Deep), Interior Empty-Cold (Tight-Weak-Deep). Pain.
15. Wiry 弦 Xián	Taut, long, and forceful whether pressed or not; maintains a straight line beneath the fingers. <i>Classical image:</i> A spring stretched on a musical instrument.	Liver disharmony. Pain. Phlegm.
16. Slowed-Down/ Moderate 緩 Huǎn	Beats four times per breath — faster than the Slow pulse.	Generally a healthy pulse — the one qualitatively normal entry on this list.
17. Hollow 芤 Kōu	Wide and soft when touched lightly, hollow when seeking; edges clearly felt under pressure but the center is obscure. <i>Classical image:</i> A scallion or stalk — has an edge but is hollow inside.	Hemorrhage. Forthcoming hemorrhage (Hollow+Rapid). Yin Deficiency.
18. Leather 革 Gé	Large, thick, and wide; rapid and forceful when lightly touched, but hollow when pressed deeply. <i>Classical image:</i> The surface of a drum — thick on top, hollow beneath.	Severe Kidney-Essence Deficiency. Severe Kidney-Yin Deficiency.
19. Firm/Confined 牢 Láo	Deep, taut, long, wide, and forceful; deeply located, cannot be felt with light or moderate pressure.	Blood Stasis. Interior Cold (Firm+Slow). Pain.
20. Soggy 濡 Rú	Superficial, thin, and forceless; grows weaker and weaker the more it's pressed. <i>Classical image:</i> A strand of cotton floating on water; a bubble that bursts under light pressure.	Dampness with Qi Deficiency. Yin Deficiency. Essence Deficiency.
21. Weak/Frail 弱 Ruò	Forceless, deep, and thin; only felt at the deep level by pressing; lost entirely with more pressure.	Yang Deficiency. Blood Deficiency.
22. Scattered 散 Sǎn	Wide, superficial, forceless, and slack when touched lightly; indistinct; disappears under pressure, rate hard to count. <i>Classical image:</i> Flower blossoms lightly scattering to the ground.	Severe Qi and Blood Deficiency. Severe Kidney-Qi Deficiency. A serious condition.

23. Hidden 伏 Fú	Cannot be felt touching, seeking, or pressing — only felt pressing very heavily to the bone. <i>Classical image:</i> Pulsation seeming to come from beneath the bone and tendons.	Severe Yang Deficiency.
24. Moving 動 Dòng	Rapid and slippery; forceful specifically at the middle portion. <i>Classical image:</i> A bean bouncing in the vessel — well-defined peaks, no head or tail.	Shock, anxiety, fright. Pain.
25. Hasty 促 Cù	Beats rapidly with irregular, random stops. <i>Classical image:</i> Agitated and urgent, like a man running and suddenly falling down.	Severe Heat. Deficiency of Heart-Qi. Heart-Fire.
26. Knotted 結 Jié	Slow with irregularly missed beats; the pulse resumes immediately after the missed beat.	Cold with Heart-Yang Deficiency.
27. Intermittent 代 Dài	Pauses at <i>regular</i> intervals; resumes after the pause.	Serious problem in the internal organs. Heart problem (Western medical sense).
28. Hurried/Racing 疾 Jí	Rapid, beating more than seven times per breath (140–180bpm) — an extreme version of Rapid; rhythm stays regular.	Fire exhausting Yin.

Two Traps Hiding in This List

Slowed-Down/Moderate (Huǎn) is the one qualitatively "good" pulse on this list — four beats per breath, generally healthy, easy to mistake for Slow if you're not counting carefully. **Knotted vs. Intermittent vs. Hasty** are all irregular-rhythm pulses but differ by speed and by whether the pause is random (Knotted=slow+random stop, Hasty=rapid+random stop) or regular (Intermittent=pauses at fixed intervals — the most serious of the three, flagging a real internal-organ problem).

10 Complete Combinations Reference

Sections 6–7 introduced the combinatorial logic with a representative sample of combinations for teaching purposes. This is the **full** combination list for every quality Schrier's slides give combinations for, sourced directly from the deck — use this table, not Sections 6–7, as the authoritative list for quiz prep.

Quality	Full combination list
Floating (Fú)	· Floating-Weak: External-Empty condition — weak Wei Qi failing to react properly to external pathogenic invasion · Floating-Empty: Summer-Heat invasion (more strongly floating), or Yin Deficiency (comparatively less floating) — degree of float is the differentiator · Floating-Slippery: invasion of Wind-Damp, or Wind complicated by Phlegm

Deep (Chén)	· Deep-Weak: Yang Deficiency · Deep-Wiry: Blood Stasis, or a Chong Mai issue · Deep-Wiry-Slow: Blood stasis due to Cold, or Cold stagnating in the Liver channel · Deep-Slow: interior Cold · Deep-Rapid: interior Heat · Deep-Soggy-Slow: Dampness in the Interior
Slow (Chí)	· Slow-Floating: exterior Wind-Cold with a prevalence of Wind · Slow-Deep: interior Cold · Slow-Slippery: Cold-Phlegm or Damp-Cold · Slow-Choppy: Blood Deficiency and interior Cold · Slow-Weak: Empty-Cold with Yang Deficiency · Slow-Full: internal Cold, often chronic painful conditions · Slow-Wiry: stagnation of Cold in the Liver channel
Rapid (Shuò)	· Rapid-Full: Full-Heat from Excess of Yang · Rapid-Empty: Empty-Heat from Deficiency of Yin · Rapid-Fine / Rapid-Floating: External Wind-Heat · Rapid-Deep: interior Heat · Rapid-Overflowing: Full-Heat · Rapid-Wiry: Liver-Fire · Rapid-Slippery: Phlegm-Heat
Full (Shí)	· Full-Rapid: Full-Heat · Full-Slow: Full-Cold · Full-Tight: Full-Cold · Full-Slippery: Phlegm · Full-Long: Heat · Full-Wiry: Full Liver pattern
Slippery (Huá)	· Slippery-Floating: Wind-Phlegm · Slippery-Rapid: Damp-Heat or Phlegm-Heat · Slippery-Slow: Damp-Cold or Cold-Phlegm · Slippery-Big: Phlegm-Heat with predominance of Heat · Slippery-Short: Dampness or Phlegm with Qi Deficiency · Slippery-Weak: Dampness or Phlegm with Qi Deficiency
Choppy (Sè)	· Choppy-Weak: depletion of both Qi and Blood · Choppy-Minute: extreme Blood Deficiency · Choppy-Soggy: Qi and Blood Deficiency with some Dampness · Choppy-Floating: invasion of external Wind · Choppy-Deep: longstanding Blood Deficiency · Choppy-Full: Blood Stasis (potentially) · Choppy-Knotted: Blood Stasis deriving from Yang Deficiency and internal Cold · Choppy-Fine: Deficiency of Body Fluids
Long (Cháng)	· Long-Rapid: internal Heat · Long-Floating: external Wind-Heat · Long-Deep: internal Heat · Long-Wiry: Liver disharmony (usually excess) · Long-Slippery: Phlegm-Heat · Long-Overflowing: excess of Yang and Heat

Short (Duǎn)	· Short-Floating: Lung-Qi Deficiency · Short-Choppy: Heart-Qi Deficiency · Short-Rapid: severe Heart-Blood Deficiency with Empty-Heat · Short-Slow: Yang Deficiency with internal Cold
Overflowing/Flooding (Hóng)	· Overflowing-Floating: Wind-Heat invasion with intense Heat, can progress quickly to the Qi level · Overflowing-Deep: internal Heat · Overflowing-Slippery: Phlegm-Heat with prevalence of Heat · Overflowing-Soft: Yin Deficiency with exhaustion of Body Fluids and Empty-Heat · Overflowing-Tight: Painful Obstruction Syndrome (Bi Syndrome)
Fine/Thin (Xì)	· Fine-Rapid: Yin Deficiency with Empty-Heat · Fine-Wiry: Liver-Qi Stagnation · Fine-Tight: Empty-Cold with underlying Blood Deficiency · Fine-Choppy: severe Blood or Essence Deficiency · Fine-Weak: severe Blood Deficiency
Minute/Faint (Wēi)	· Minute-Rapid: Yin Deficiency with Empty-Heat · Minute-Slow: severe Qi and Blood Deficiency with internal Cold · Minute-Choppy: exhaustion of Blood · Minute-Soggy: Qi and Blood Deficiency with Dampness · Minute-Wiry: Liver-Yang rising with underlying Blood Deficiency
Tight (Jǐn)	· Tight-Floating: Wind-Cold invasion, prevalence of Cold · Tight-Deep: internal Full-Cold · Tight-Choppy: Cold Bi Syndrome · Tight-Slippery: Cold-Phlegm · Tight-Firm: Blood Stasis from Cold
Wiry (Xián)	· Wiry-Rapid: Liver-Fire · Wiry-Slow: stagnation of Cold in the Liver channel · Wiry-Long: Liver-Qi Stagnation or Liver-Blood Stasis · Wiry-Tight: Blood Stasis · Wiry-Big: Liver-Fire or Liver-Yang rising · Wiry-Overflowing: Liver-Fire
Slowed-Down/ Moderate (Huǎn)	· Slowed-Down-Soggy: Dampness with Spleen-Qi Deficiency · Slowed-Down-Slippery: Cold-Dampness · Slowed-Down-Fine: Qi and Blood Deficiency · Slowed-Down-Floating: weakness of the Wei Qi · Slowed-Down-Deep: weakness of the Ying Qi · Slowed-Down-Choppy: Blood Deficiency
Hollow (Kōu)	· Hollow-Rapid: Yin Deficiency with Empty-Heat, or impending hemorrhage · Hollow-Empty-Soggy: loss of Essence or depletion of Blood · Hollow-Knotted: Yang Deficiency or Blood Stasis
Leather (Gé)	· Leather-Rapid: severe Deficiency of Yin with Empty-Heat · Leather-Slow: severe Blood Deficiency with Blood Stasis · Leather-Choppy: severe Blood Deficiency

Firm/Confined (Láo)	· Firm-Tight: internal Cold · Firm-Slow: internal Cold and Blood Stasis · Firm-Choppy: Blood Stasis
Soggy (Rú)	· Soggy-Fine: severe Spleen-Qi Deficiency · Soggy-Choppy: exhaustion of Blood · Soggy-Floating: Deficiency of Wei Qi with Dampness · Soggy-Wiry: Dampness with Liver-Qi Stagnation · Soggy-Rapid: Damp-Heat
Weak/Frail (Ruò)	· Weak-Choppy: Blood Deficiency · Weak-Fine: severe Blood Deficiency · Weak-Slow: Yang Deficiency · Weak-Soggy: Yang Deficiency with some Dampness
Scattered (Sàn)	· Scattered-Slow: severe Qi and Yang Deficiency · Scattered-Rapid: severe Blood Deficiency with Empty-Heat · Scattered-Floating: severe Deficiency of the Yuan Qi with floating of Yang, or with Yin Fire
Hidden (Fú)	· Hidden-Full: Qi and Blood stagnation; retention of food; internal Cold; internal Heat · Hidden-Empty: severe Yang Deficiency with internal Cold · Hidden-Rapid: interior Heat · Hidden-Slow: interior Cold
Moving (Dòng)	· Moving-Rapid: Heat · Moving-Slippery: Deficiency of Qi and Phlegm · Moving-Empty: severe depletion of Blood · Moving-Weak: Shock
Hasty (Cù)	· Hasty-Overflowing: Heat in the Stomach · Hasty-Slippery: retention of Phlegm-Heat · Hasty-Wiry: Liver-Fire and Heart-Fire · Hasty-Slippery-Wiry: Heart-Fire, Liver-Fire, and Phlegm-Heat · Hasty-Fine without strength: collapse of Heart-Qi
Knotted (Jié)	· Knotted-Choppy: Blood Stasis · Knotted-Slippery: retention of Phlegm · Knotted-Deep: accumulation in the Interior · Knotted-Wiry: disease of the arteries, or hypertension
Hurried/Racing (Jí)	· Hurried-Floating: severe Empty-Heat from Yin Deficiency · Hurried-Floating-Empty: severe Yin Deficiency · Hurried-Slippery: Empty-Heat from Yin Deficiency with Phlegm · Hurried-Overflowing: Empty-Heat arising from Yin Deficiency

Empty/Deficient, Intermittent, and Hidden(partially) had no dedicated combinations slide in the deck — only base indications, already listed in Section 9.

11 The Pulse Quality Comparison Chart

Schrier's slides include a dedicated visual — the one he specifically tells students to print out and bring to class — that groups all 28 qualities under the four dimensions used to describe any pulse: **Speed/Rate**, **Strength/Arrival**, **Rhythm**, and **Position (Superficial/Floating vs. Deep/Submerged)**. Each dimension has two poles, and qualities cluster toward whichever pole they primarily express.

Sourcing Note

This reconstruction comes from OCR text-position data on the original slide, not a direct visual read — I could not get a clean image render of this chart this session. The grouping below is accurate to what's on the slide; the exact visual layout (spacing, connecting lines, precise spectrum position of each quality) is not reproduced. Cross-check against the printed slide itself before relying on this for exact placement.

Dimension	Pole 1	Pole 2
Speed/Rate	<i>Rapid/Fast side:</i> Rapid, Moving, Hurried/Racing, Slippery, Overflowing/Flooding	<i>Slow side:</i> Slow, Choppy, Moderate/Slowed-Down, Minute/Faint, Fine/Thin, Weak/Frail, Short
Strength/Arrival	<i>Excess side:</i> Tight/Tense, Full/Excess, Overflowing/Flooding/Surging	<i>Deficiency/Empty side:</i> Deficiency, Minute, Weak/Frail
Rhythm	<i>Irregular cluster:</i> Knotted, Hasty, Intermittent	<i>Noted adjacent on the slide:</i> Floating, Soggy
Position	<i>Superficial/Floating side:</i> Leather/Drumskin, Scattered, Hollow/Scallion	<i>Deep/Submerged side:</i> Deep, Firm/Confined, Hidden

Why This Chart Is Worth Having

Several qualities appear on more than one axis — Weak/Frail shows up under both Speed and Strength, Overflowing/Flooding under both Speed and Strength as well. That's not an inconsistency in the chart; it's the point. A quality like Weak/Frail is defined by *both* a slow-leaning quality *and* a deficient-leaning quality at once — the chart is showing you that pulse qualities are compound descriptions along multiple simultaneous dimensions, which is the same "Lego" combinatorial logic from Section 6, just visualized instead of tabulated.

PART II · Complete Reference

1 Pulse Positions — 寸關尺

Position	Left	Right	Burner
寸 Cùn (1st, Front)	Hand Taiyang / SI · Hand Shaoyin / HT	Hand Yangming / LI · Hand Taiyin / LU	Upper Burner
關 Guān (2nd, Middle)	Foot Shaoyang / GB · Foot Jueyin / LR	Foot Yangming / ST · Foot Taiyin / SP	Middle Burner
尺 Chǐ (3rd, Rear)	Foot Taiyang / BL · Foot Shaoyin / KD	Hand Shaoyang / TB · Hand Jueyin / PC	Lower Burner

Note on Schools

Some modern shorthand simplifies the two Chǐ positions to "Kidney-Yin / Kidney-Yang." Schrier teaches the full Nan Jing organ pairings above — use those.

2 The Five Finger Movements

#	Movement	Checks for
1	Lifting	Superficial level — light touch
2	Pressing	Deep level — firm touch to the bone
3	Searching	Rate
4	Pushing	Shape
5	Rolling	Length

3 Components of a Normal Pulse — the 8 Parameters

Component	Normal state
Width	Diameter of the vessel, dependent on Qi and Blood. Should be moderate — neither too thin nor too wide.
Depth	Where the pulse is felt strongest. Normal is the middle depth — Qi and Blood flow freely interior ↔ exterior.
Rate	4–5 beats per respiratory cycle, or 60–90bpm. Rate reflects the state of the Yang.
Rhythm	Pattern of beats. Normal is regular — closely tied to Heart Qi, which commands Blood through the vessels.
Strength	Beating force beneath the finger. Moderate — gentle and soft, neither forceful nor forceless.
Length	Length across the three positions. Felt at all positions, neither too long nor short. Qi governs length.
Tension	Neither too taut nor too slack. Predominantly a manifestation of the condition of Qi and Blood.
Shape	Wave or contour — rising and falling as blood flows. Ascending and descending should be about equal.

4 胃神根 — The Three Qualities

Character	Quality	Meaning
胃 Wèi	Stomach-Qi	Flowing, gentle, harmonious movement
神 Shén	Spirit	Strong, rhythmical — vitality present in the beat
根 Gēn	Root	Depth felt at Chǐ — Kidney Qi intact under firm pressure

5 Seasonal Pulses

Season	Pulse	Dates	Why
Spring	Wiry 弦	Feb 4 – May 4	Yang Qi wants to rise; Cold-Yin nature still obstructs release — shouldn't be too wiry
Summer	Flooding 洪	May 5 – Aug 6	Yang Qi uninhibited, spreading freely, at its peak
Autumn	Hair-like / Floating 毛	Aug 7 – Nov 6	Yang Qi desiring to be restrained and collected — "dainty, like elm pods falling"
Winter	Stone / Deep 石	Nov 7 – Feb 3	Yang Qi gathered and stored; sinks like a stone

6 The 28 Pulse Qualities — Full Reference

Quality	Sensation & Classical Image	Indications
1. Floating 浮 Fú	Felt distinctly at the superficial depth; force decreases slightly deeper but never feels empty. <i>Classical:</i> Wood floating on the surface of water; a subtle breeze across a bird's back; scallion leaves rolled lightly between the fingers.	Invasion of external Wind. Yin Deficiency (interior condition presenting superficially).
2. Deep / Submerged 沉 Chén	Located deep next to the bone; barely felt with light or moderate pressure; only felt pressing to bone and tendons. <i>Classical:</i> A stone thrown into a well, sinking to the bottom.	Pathogenic factor in the Interior (Deep-Full). Yang Deficiency (Deep-Weak).
3. Slow 遲 Chí	Fewer than three beats per breath, about 40–60bpm, regular rhythm.	Cold pattern. Empty-Cold (Slow+Weak). Full-Cold (Slow+Full).
4. Rapid 數 Shuò	Over five beats per breath, 90–140bpm, regular rhythm.	Heat pattern. Full-Heat (Rapid+Full). Empty-Heat (Rapid+Floating-Empty).
5. Empty / Deficient 虛 Xū	Wide but without strength; forceless at all three positions under light or heavy pressure; a sense of emptiness. <i>Classical:</i> A water balloon — big on the outside, nothing solid when you press in.	Qi Deficiency.
6. Full / Excessive 實 Shí	Felt at all three positions and all three depths; long, wide, and forceful.	Full pattern. Full-Heat (Full+Rapid). Full-Cold (Full+Slow).
7. Slippery 滑 Huá	Feels quick though not rapid; slippery and smooth; rounded inside the vessel, as if oily. <i>Classical:</i> A pearl rolling on a plate; a raindrop gliding off a leaf.	Phlegm. Dampness. Retention of food. Pregnancy.
8. Choppy 澀 Sè	Seems slow and is not smooth; vessel is thin; rhythm irregular in force and fullness; feels rough, jagged. <i>Classical:</i> Scraping bamboo with a small knife.	Blood Deficiency. Exhaustion of Body Fluids.
9. Long 長 Cháng	Smooth and straight; felt extending distally and proximally beyond the usual positions.	Heat patterns.
10. Short 短 Duǎn	Cannot be felt in all three positions; absent at the distal and/or proximal position.	Severe Qi Deficiency. Stomach-Qi Deficiency.
11. Overflowing / Flooding 洪 Hóng	Wide and forceful, though it fades gently; vessel wide and thick. <i>Classical:</i> Ocean waves that come strongly but leave calmly.	Full-Heat. Empty-Heat (Overflowing-Empty).

Quality	Sensation & Classical Image	Indications
12. Fine / Thin 細 <i>Xī</i>	Very thin, like a fine thread, but distinct and clear without interruption of rhythm under pressure. <i>Classical:</i> Angel-hair pasta rather than a thick udon noodle.	Blood Deficiency. Dampness with severe Qi Deficiency.
13. Minute / Faint 微 <i>Wēi</i>	Extremely thin and forceless; seems felt, then not felt; pressing deeply the vessel becomes vague. <i>Classical:</i> Now you see it, now you don't.	Severe deficiency of Qi and Blood.
14. Tight 緊 <i>Jǐn</i>	Rises and falls with strength, vibrating left and right; feels twisted like a thick rope. <i>Classical:</i> Fingers on a highly stretched twisted cord, snapping back and forth.	Cold — Exterior (Tight-Floating), Interior Full-Cold (Tight-Full-Deep), Interior Empty-Cold (Tight-Weak-Deep). Pain.
15. Wiry 弦 <i>Xián</i>	Taut, long, and forceful whether pressed or not; maintains a straight line beneath the fingers. <i>Classical:</i> A spring stretched on a musical instrument; a guitar string about to snap.	Liver disharmony. Pain. Phlegm.
16. Slowed-Down / Moderate 緩 <i>Huǎn</i>	Beats four times per breath — faster than the Slow pulse.	Generally a HEALTHY pulse — the one qualitatively normal entry on this list.
17. Hollow 芤 <i>Kōu</i>	Wide and soft when touched lightly, hollow when seeking; edges clearly felt under pressure but the center is obscure. <i>Classical:</i> A scallion stalk — has an edge but is hollow inside.	Hemorrhage. Forthcoming hemorrhage (Hollow+Rapid). Yin Deficiency.
18. Leather 革 <i>Gé</i>	Large, thick, and wide; rapid and forceful when lightly touched, but hollow when pressed deeply. <i>Classical:</i> The surface of a drum — thick on top, hollow beneath.	Severe Kidney-Essence Deficiency. Severe Kidney-Yin Deficiency.
19. Firm / Confined 牢 <i>Láo</i>	Deep, taut, long, wide, and forceful; deeply located, cannot be felt with light or moderate pressure.	Blood Stasis. Interior Cold (Firm+Slow). Pain.
20. Soggy 濡 <i>Rú</i>	Superficial, thin, and forceless; grows weaker and weaker the more it is pressed. <i>Classical:</i> A strand of cotton floating on water; a bubble that bursts under light pressure.	Dampness with Qi Deficiency. Yin Deficiency. Essence Deficiency.
21. Weak / Frail 弱 <i>Ruò</i>	Forceless, deep, and thin; only felt at the deep level by pressing; lost entirely with more pressure.	Yang Deficiency. Blood Deficiency.
22. Scattered 散 <i>Sàn</i>	Wide, superficial, forceless, and slack when touched lightly; indistinct; disappears under pressure, rate hard to count. <i>Classical:</i> Flower blossoms lightly scattering to the ground.	Severe Qi and Blood Deficiency. Severe Kidney-Qi Deficiency. A serious condition.

Quality	Sensation & Classical Image	Indications
23. Hidden 伏 Fú	Cannot be felt touching, seeking, or pressing — only felt pressing very heavily to the bone. <i>Classical:</i> Pulsation seeming to come from beneath the bone and tendons.	Severe Yang Deficiency.
24. Moving 動 Dòng	Rapid and slippery; forceful specifically at the middle portion. <i>Classical:</i> A bean bouncing in the vessel — well-defined peaks, no head or tail.	Shock, anxiety, fright. Pain.
25. Hasty 促 Cù	Beats rapidly with irregular, random stops. <i>Classical:</i> Agitated and urgent, like a man running and suddenly falling down.	Severe Heat. Deficiency of Heart-Qi. Heart-Fire.
26. Knotted 結 Jié	Slow with irregularly missed beats; the pulse resumes immediately after the missed beat.	Cold with Heart-Yang Deficiency.
27. Intermittent 代 Dài	Pauses at REGULAR intervals; resumes after the pause.	Serious problem in the internal organs. Heart problem (Western medical sense).
28. Hurried / Racing 疾 Jí	Rapid, beating more than seven times per breath (140–180bpm) — an extreme version of Rapid; rhythm stays regular.	Fire exhausting Yin.

7 Complete Combinations Reference

Quality	Full combination list
Floating (Fú)	· Floating-Weak: External-Empty — weak Wei Qi failing to react properly to external pathogenic invasion · Floating-Empty: Summer-Heat invasion (floats MORE) vs. Yin Deficiency (floats less) — degree is the differentiator · Floating-Slippery: invasion of Wind-Damp, or Wind complicated by Phlegm
Deep / Submerged (Chén)	· Deep-Weak: Yang Deficiency · Deep-Wiry: Blood Stasis, or a Chong Mai issue · Deep-Wiry-Slow: Blood stasis due to Cold, or Cold stagnating in the Liver channel · Deep-Slow: interior Cold · Deep-Rapid: interior Heat · Deep-Soggy-Slow: Dampness in the Interior

Quality	Full combination list
Slow (Chí)	· Slow-Floating: exterior Wind-Cold with prevalence of Wind · Slow-Deep: interior Cold · Slow-Slippery: Cold-Phlegm or Damp-Cold · Slow-Choppy: Blood Deficiency and interior Cold · Slow-Weak: Empty-Cold with Yang Deficiency · Slow-Full: internal Cold, often chronic painful conditions · Slow-Wiry: stagnation of Cold in the Liver channel
Rapid (Shuò)	· Rapid-Full: Full-Heat from Excess of Yang · Rapid-Empty: Empty-Heat from Deficiency of Yin · Rapid-Fine / Rapid-Floating: External Wind-Heat · Rapid-Deep: interior Heat · Rapid-Overflowing: Full-Heat · Rapid-Wiry: Liver-Fire · Rapid-Slippery: Phlegm-Heat
Full / Excessive (Shí)	· Full-Rapid: Full-Heat · Full-Slow: Full-Cold · Full-Tight: Full-Cold · Full-Slippery: Phlegm · Full-Long: Heat · Full-Wiry: Full Liver pattern
Slippery (Huá)	· Slippery-Floating: Wind-Phlegm · Slippery-Rapid: Damp-Heat or Phlegm-Heat · Slippery-Slow: Damp-Cold or Cold-Phlegm · Slippery-Big: Phlegm-Heat with predominance of Heat · Slippery-Short: Dampness or Phlegm with Qi Deficiency · Slippery-Weak: Dampness or Phlegm with Qi Deficiency
Choppy (Sè)	· Choppy-Weak: depletion of both Qi and Blood · Choppy-Minute: extreme Blood Deficiency · Choppy-Soggy: Qi and Blood Deficiency with some Dampness · Choppy-Floating: invasion of external Wind · Choppy-Deep: longstanding Blood Deficiency · Choppy-Full: Blood Stasis (potentially) · Choppy-Knotted: Blood Stasis from Yang Deficiency and internal Cold · Choppy-Fine: Deficiency of Body Fluids
Long (Cháng)	· Long-Rapid: internal Heat · Long-Floating: external Wind-Heat · Long-Deep: internal Heat · Long-Wiry: Liver disharmony (usually excess) · Long-Slippery: Phlegm-Heat · Long-Overflowing: excess of Yang and Heat
Short (Duǎn)	· Short-Floating: Lung-Qi Deficiency · Short-Choppy: Heart-Qi Deficiency · Short-Rapid: severe Heart-Blood Deficiency with Empty-Heat · Short-Slow: Yang Deficiency with internal Cold

Quality	Full combination list
Overflowing / Flooding (Hóng)	· Overflowing-Floating: Wind-Heat with intense Heat — can progress quickly to the Qi level · Overflowing-Deep: internal Heat · Overflowing-Slippery: Phlegm-Heat with prevalence of Heat · Overflowing-Soft: Yin Deficiency with exhaustion of Body Fluids and Empty-Heat · Overflowing-Tight: Painful Obstruction (Bi) Syndrome
Fine / Thin (Xì)	· Fine-Rapid: Yin Deficiency with Empty-Heat · Fine-Wiry: Liver-Qi Stagnation · Fine-Tight: Empty-Cold with underlying Blood Deficiency · Fine-Choppy: severe Blood or Essence Deficiency · Fine-Weak: severe Blood Deficiency
Minute / Faint (Wēi)	· Minute-Rapid: Yin Deficiency with Empty-Heat · Minute-Slow: severe Qi & Blood Deficiency with internal Cold · Minute-Choppy: exhaustion of Blood · Minute-Soggy: Qi & Blood Deficiency with Dampness · Minute-Wiry: Liver-Yang rising with underlying Blood Deficiency
Tight (Jǐn)	· Tight-Floating: Wind-Cold invasion, prevalence of Cold · Tight-Deep: internal Full-Cold · Tight-Choppy: Cold Bi Syndrome · Tight-Slippery: Cold-Phlegm · Tight-Firm: Blood Stasis from Cold
Wiry (Xián)	· Wiry-Rapid: Liver-Fire · Wiry-Slow: stagnation of Cold in the Liver channel · Wiry-Long: Liver-Qi Stagnation or Liver-Blood Stasis · Wiry-Tight: Blood Stasis · Wiry-Big: Liver-Fire or Liver-Yang rising · Wiry-Overflowing: Liver-Fire
Slowed-Down / Moderate (Huǎn)	· Slowed-Down-Soggy: Dampness with Spleen-Qi Deficiency · Slowed-Down-Slippery: Cold-Dampness · Slowed-Down-Fine: Qi and Blood Deficiency · Slowed-Down-Floating: weakness of the Wei Qi · Slowed-Down-Deep: weakness of the Ying Qi · Slowed-Down-Choppy: Blood Deficiency
Hollow (Kōu)	· Hollow-Rapid: Yin Deficiency with Empty-Heat, or impending hemorrhage · Hollow-Empty-Soggy: loss of Essence or depletion of Blood · Hollow-Knotted: Yang Deficiency or Blood Stasis
Leather (Gé)	· Leather-Rapid: severe Deficiency of Yin with Empty-Heat · Leather-Slow: severe Blood Deficiency with Blood Stasis · Leather-Choppy: severe Blood Deficiency
Firm / Confined (Láo)	· Firm-Tight: internal Cold · Firm-Slow: internal Cold and Blood Stasis · Firm-Choppy: Blood Stasis

Quality	Full combination list
Soggy (Rú)	· Soggy-Fine: severe Spleen-Qi Deficiency · Soggy-Choppy: exhaustion of Blood · Soggy-Floating: Deficiency of Wei Qi with Dampness · Soggy-Wiry: Dampness with Liver-Qi Stagnation · Soggy-Rapid: Damp-Heat
Weak / Frail (Ruò)	· Weak-Choppy: Blood Deficiency · Weak-Fine: severe Blood Deficiency · Weak-Slow: Yang Deficiency · Weak-Soggy: Yang Deficiency with some Dampness
Scattered (Sàn)	· Scattered-Slow: severe Qi and Yang Deficiency · Scattered-Rapid: severe Blood Deficiency with Empty-Heat · Scattered-Floating: severe Deficiency of Yuan Qi with floating of Yang, or with Yin Fire
Hidden (Fú)	· Hidden-Full: Qi and Blood stagnation; retention of food; internal Cold; internal Heat · Hidden-Empty: severe Yang Deficiency with internal Cold · Hidden-Rapid: interior Heat · Hidden-Slow: interior Cold
Moving (Dòng)	· Moving-Rapid: Heat · Moving-Slippery: Deficiency of Qi and Phlegm · Moving-Empty: severe depletion of Blood · Moving-Weak: Shock
Hasty (Cù)	· Hasty-Overflowing: Heat in the Stomach · Hasty-Slippery: retention of Phlegm-Heat · Hasty-Wiry: Liver-Fire and Heart-Fire · Hasty-Slippery-Wiry: Heart-Fire, Liver-Fire, and Phlegm-Heat · Hasty-Fine without strength: collapse of Heart-Qi
Knotted (Jié)	· Knotted-Choppy: Blood Stasis · Knotted-Slippery: retention of Phlegm · Knotted-Deep: accumulation in the Interior · Knotted-Wiry: disease of the arteries, or hypertension
Hurried / Racing (Jí)	· Hurried-Floating: severe Empty-Heat from Yin Deficiency · Hurried-Floating-Empty: severe Yin Deficiency · Hurried-Slippery: Empty-Heat from Yin Deficiency with Phlegm · Hurried-Overflowing: Empty-Heat arising from Yin Deficiency

8 Pulse Quality Comparison — Schrier's Chart

Dimension	Pole 1	Pole 2
Speed / Rate	Rapid/Fast: Rapid, Moving, Hurried/Racing, Slippery	Slow: Slow, Choppy, Moderate/Slowed-Down
Form — Shape	Wiry, Slippery, Choppy, Tight	—

Dimension	Pole 1	Pole 2
Form — Size/Width	Big/Large, Minute/Faint, Fine/Thin, Overflowing/Flooding/Surging, Weak/Frail	—
Form — Length	Short	Long
Strength / Arrival	Excess: Overflowing/Flooding/Surging, Tight/Tense, Full/Excess	Deficiency: Deficiency/Empty, Minute, Weak/Frail
Rhythm	Knotted, Hasty, Intermittent	—
Depth	Superficial/Floating: Floating, Soggy, Leather/Drumskin, Scattered, Hollow/Scallion	Deep: Deep/Submerged, Firm/Confined, Hidden

PART III · Method, Technique & Context

Why this document exists

The main Study Guide covers the 28 qualities and their combinations. This covers everything **around** them that Schrier taught in lecture but that isn't on the pulse-quality slides — the depths, the physical technique, the channel-flow material, and the classical frameworks he referenced. Several of these he flagged as "might come up later."

1 The Three Depths — Not the Same as the Three Positions

This is the distinction students most often collapse

Positions = Cùn / Guān / Chǐ, along the artery (front to back). **Depths** = superficial / middle / deep, pressing downward. You assess all three depths *at each* of the three positions.

Depth	How to find it	Maciocia's reading	Alternative teaching
Superficial	Light contact — just touch the skin	Qi	Yang organs (Yang is up)
Middle	The hard one. Press to bone, obliterate the pulse, then release <i>slowly</i> until something appears	Blood	The balance between the two
Deep	Press all the way to bone level	Yin organs	Yin organs (Yin is deep)

Schrier's practice instruction

"Start here, press gently, feel that position. Press all the way down — don't think about the middle. Feel that position. Obliterate the pulse, lift up, find that sensation, then slowly go up, slowly go down. Wax on, wax off." The superficial and deep are the easy two; the middle emerges from moving between them. He is explicit that skipping the middle because it's hard is how you build a permanent bad habit.

A legitimate simplification

The Worsley system uses **no pulse qualities at all** — only positions and depths, assessed for relative strength. That is a real lineage, not a shortcut. But learn the full method first; choose to simplify later.

2 Physical Technique

Element	How
Arm position	Horizontal. Not above heart level. Minimise bends and crooks in the arm — Qi moves poorly through kinks (the qigong principle: open, rounded, not robotic).
Which hand	Your right hand reads their left wrist; your left reads their right. If you're beside a supine patient, walk around rather than contorting. Same-side facing reverses it.
Finger placement	Index = Cùn, middle finger always on Guān , ring = Chǐ. Place all three simultaneously first for an overall impression, then assess each position individually.
Spacing	Adjust to the patient's forearm length — tighter on a short arm, more spread on a long one.
Fingernails	Long nails are workable but you must use the pads above the fingertip, not the tips. Don't stab.
Calluses	Guitar/guqin calluses do change sensitivity. They don't disqualify you — physicians have always played instruments — just know your own hands.
Pulse pillow	Various types. Support the wrist so the patient isn't holding tension.
Position	Seated facing you, or supine. Both are valid; practise both.

Fingers are NOT held still — a common misconception

You keep the fingers still **only** when counting rate (you can't judge fast vs. slow while moving). Otherwise you are actively moving: **press and lift** for depth · **roll left and right** for length · **push medially and laterally** for shape and slippery quality. This is what the five finger movements actually are in the hand.

Counting the rate against your own breath

There is no prescribed second-count. Schrier's answer to "how many seconds is one breath?" was essentially: breathe naturally, don't engineer it. If you force slow or force fast you corrupt your own reference. Watch the patient's chest, breathe normally, count. Note also that patients often try to 'help' by controlling their breathing — which shifts the pulse. Don't invite it.

3 Timing and Context — What Moves a Pulse That Isn't Pathology

Factor	Effect
Recent exertion	Stairs, rushing, running late → rapid pulse that is not heat. Let them settle before you read anything.
Room temperature	A hot or cold room shifts the pulse. Ask for the fan if needed.
Time of day	Organ-clock timing means the corresponding organ's pulse reads stronger. Schrier treats the two-hour blocks as a general rule anchored to actual sunrise/sunset, not clock time — and notes daylight saving muddies it further.
Classical ideal	Historically the pulse was taken first thing in the morning before Yang rises. Almost nobody's clinic runs that way — know the ideal, work with reality.
Season	See the seasonal pulses. Normal-for-season is only normal when nothing else corroborates pathology.
Order in a rotation	In class practice the pulse will have changed by the last person to feel it. Expected — not your failure.
Diet / substances	Coffee will yellow a coating; spicy food creates transient heat. Ask before you conclude.

The two attentional instructions

Ling Shu: "Shut the door and close the windows, attune yourself directly with your patient." Yi (intention) leads Qi — divided attention degrades what you can feel.

The fishing analogy: you cannot force the pulse to arrive by trying harder. Agitation in you jerks the line. Sit, be patient, clear the mind. This is a technical instruction, not a mood.

4 Two Position Maps — and Why They Differ

	Nan Jing 難經	Lǐ Shìzhēn · Bīn Hú Mài Xué
Left	Cùn = SI/HT · Guān = GB/LR · Chǐ = BL/KD	Cùn = HT · Guān = LR · Chǐ = KI
Right	Cùn = LI/LU · Guān = ST/SP · Chǐ = TB/PC	Cùn = LU · Guān = SP · Chǐ = KI (Mingmen)
Emphasis	Full twelve-channel mapping — channel theory, Qi flow	Simpler organ mapping
Leans toward	Acupuncture	Herbal practice

How to hold two authoritative maps that disagree

Schrier's framing: it's like buying a map of Virginia from 2015, 2020, and 2026. Same territory, some roads moved. The move is not to pick a winner — it's to **ask which system a claim comes from**. "Where are you getting this? Which text? Whose lineage?" A disagreement you can source is a difference of school; one you can't source is a red flag. Some lineages (Li Shizhen among them) barely weight the Yang organs at all.

The nine regions — the older scheme

Before radial-only standardisation, palpation used nine locations: three on the head, three on the hands, three on the feet/legs. Su Wen describes these. Alternate leg sites persist (LR-10, SP-11) but are largely displaced by radial practice — partly for the practical reason that palpating a patient's inner thigh carries modesty considerations the wrist does not. Three recurs throughout: three positions, three burners, three depths, Jing/Qi/Shen, Heaven/Earth/Humanity, the three-legged *dīng* vessel — stability through triplicity.

5 Channel Flow and Pulse Blocks

The Ying cycle — the order to check

LU → LI → ST → SP → HT → SI → BL → KD → PC → TB → GB → LR → back to LU. Twelve channels, twelve months; the cycle begins with the Lung, which corresponds to first breath and to sunrise.

What a block feels like — the hose analogy

Pinch a garden hose: water piles up full and tight before the pinch, and runs weak and empty after it. A pulse block reads the same way — one position abnormally full and tight, the adjacent one relatively empty, at the junction between channels. Treatment is to needle the **exit and entry points** of the two channels involved to reopen the passage. This is why writing your pulse positions down matters: you cannot see a block by feeling one position at a time and not recording it.

Husband–Wife block (Worsley)

Defined as the left pulse being weaker in quality and quantity than the right. Schrier flags the naming as dated, and notes the historical texts actually say the opposite for women — that the right should be stronger, right being Yin, left Yang. He is openly skeptical the rule holds. Know the term; hold the rule loosely.

6 Shang Han Lun — Six Stage Pulses

Flagged as "not tested now, may come up later"

Worth knowing because the stage names recur constantly in later coursework.

Stage	Pulse	Anchor
Tai Yang	Floating	Exterior stage
Yang Ming	Flooding, Long	The Four Bigs — big thirst, big sweat, big pulse, big fever. Maximum heat.
Shao Yang	Wiry	Alternating hot and cold — stuck between Taiyang and Yangming. (Schrier's mnemonic: "Katy Perry syndrome" — hot and cold, yes and no.)
Tai Yin	Moderate	Middle — neither fast nor slow
Shao Yin	Faint and Thin	Depletion
Jue Yin	Tendency to expire	Terminal stage

7 Climatic Factors → Pulse

Factor	Pulse	The logic
Wind	Floating	What floats? Things carried on air.
Cold	Tight	Cold contracts. You tighten up when cold. Also: deeper water is colder — deep is Yin, surface Yang.
Summer-Heat	Floating, vacuous	Heat drives outward; the vacuity is the fluid loss.
Dampness	Soggy, Slippery	Wet things are soft and slide.
Dryness	Short, Hasty, Rough	Dry things shrink and shorten. Sandpaper is rough because it's dry.
Heat	Restless, Rapid	Broken AC on a summer night — you can't settle.

PART IV · Confusion Pairs — Where Marks Are Lost

How to use this

These are the pairs that cost marks. For each, read the shared feature first, then the differentiator, then the practical test. Cover the right column and self-quiz.

Pair	Shared feature	Differentiator	Practical test
Slow 遲 vs. Slowed-Down/Moderate 緩	Both feel unhurried.	Slow = fewer than 3 beats/breath (40–60bpm) → Cold . Huǎn = 4 beats/breath, faster → generally HEALTHY .	Count. Don't eyeball it. This is the single easiest mark to lose.
Knotted 結 vs. Hasty 促	Both have random irregular stops.	Knotted = SLOW + random stops → Cold w/ Heart-Yang Def. Hasty = RAPID + random stops → Severe Heat / Heart-Fire.	Speed is the whole differentiator.

Pair	Shared feature	Differentiator	Practical test
Knotted/Hasty vs. Intermittent 代	All three are irregular.	Knotted & Hasty stop randomly . Intermittent pauses at REGULAR intervals — and is the most serious (organ/Heart problem).	Random vs. patterned. Listen for the pattern.
Fine/Thin 細 vs. Minute/Faint 微	Both are narrow.	Fine = thin but distinct and clear , rhythm uninterrupted → Blood Def. Minute = extremely thin AND forceless , fades in and out → severe Qi and Blood Def.	Fine you can follow. Minute keeps vanishing.
Hollow 芤 vs. Leather 革	Both hollow underneath.	Hollow = scallion stalk, wide/soft on light touch, centre obscure → Hemorrhage , Yin Def. Leather = drum surface, large/thick, rapid and forceful on light touch → severe Kidney-Essence/Yin Def.	Leather has force and speed at the surface; Hollow is soft.
Empty 虛 vs. Scattered 散	Both forceless.	Empty = wide, forceless at all 3 positions, keeps its shape → Qi Def. Scattered = forceless AND loses shape , disappears under pressure, rate hard to count → severe Qi/Blood/Kidney-Qi Def, serious.	If you can't count the rate, it's Scattered.
Deep 沉 vs. Hidden 伏	Both need pressure.	Deep = felt pressing to bone/tendons. Hidden = cannot be felt at all touching/seeking/pressing; only very heavy bone-level pressure.	Hidden is a degree beyond Deep — severe Yang Def.
Deep 沉 vs. Firm/Confined 牢	Both deep.	Deep can be weak or full. Firm = deep AND taut, long, wide, forceful → Blood Stasis, interior Cold, Pain.	Firm is deep with strength and tension.
Tight 緊 vs. Wiry 弦	Both taut.	Tight = vibrates left and right , twisted rope, snapping → Cold , Pain. Wiry = straight line , taut and stable, instrument string → Liver , Pain, Phlegm.	Tight moves side to side. Wiry holds a line.
Soggy 濡 vs. Weak/Frail 弱	Both soft and forceless.	Soggy = superficial , thin, forceless, weaker the more you press → Damp + Qi Def. Weak = deep , thin, forceless, only at the deep level → Yang Def, Blood Def.	Soggy floats. Weak sinks.
Floating (Exterior) vs. Floating (Yin Def)	Same position finding.	Exterior: simultaneous aversion to cold + fever, occipital stiffness, nasal congestion. Yin Def: Floating + Thin + Rapid , night sweats, five-palm heat, malar flush, dry throat worse at night.	The chills-and-fever pairing is the tell. Lose it → not Exterior.
Floating-Empty: Summer-Heat vs. Yin Def	Identical combination name.	Summer-Heat floats MORE . Yin Deficiency's Floating-Empty floats less .	Schrier states the differentiator is degree of float.

Pair	Shared feature	Differentiator	Practical test
Choppy 澀 vs. Fine 細	Both suggest Blood Deficiency.	Choppy = rough, jagged , irregular in force and fullness, scraping bamboo. Fine = smooth , just narrow.	Texture vs. width.
Overflowing 洪 vs. Full 實	Both big and forceful.	Overflowing = wide, forceful, fades gently – waves that come strong, leave calm. Full = forceful at all three depths and positions , long and wide, doesn't fade.	Overflowing has a decay. Full is uniform.
Moving 動 vs. Slippery 滑	Both round and quick.	Moving = rapid AND slippery, forceful at the middle position only , bean bouncing, no head or tail → Shock/fright, Pain. Slippery = smooth glide throughout → Phlegm/Damp/food/pregnancy.	Moving is localised and startling.

PART V · Red Flags — When to Refer

What this sheet is

Findings that change your disposition — not just your diagnosis. Several of these warrant referral or escalation rather than needling.

Pulse	How it presents	Indicates	What to do about it
Intermittent 代	Pauses at REGULAR intervals	Serious internal organ problem; Heart problem in the Western medical sense	Refer for cardiac evaluation. Do not treat as a simple TCM rhythm variant.
Scattered 散	Forceless, loses shape, disappears under pressure, rate hard to count	Severe Qi & Blood Deficiency; severe Kidney-Qi Deficiency	Explicitly flagged on Schrier's slide as a serious condition .
Hasty + Fine without strength	Rapid, irregular stops, thin, no force	Collapse of Heart-Qi	Emergency-level finding. Escalate.
Hidden 伏	Only felt pressing very heavily to bone	Severe Yang Deficiency	Also seen in extreme pain/collapse states. Assess for shock.
Hurried/Racing 疾	Over 7 beats/breath, 140–180bpm	Fire exhausting Yin	Extreme. Check hydration, fever, thyroid; escalate if sustained.
Leather 革	Drum surface — thick above, hollow beneath	Severe Kidney-Essence / Kidney-Yin Deficiency	Often follows major blood/fluid loss or chronic depletion.
Hollow 芤	Scallion — edges felt, centre obscure	Hemorrhage, or FORTHCOMING hemorrhage if Hollow + Rapid	Hollow+Rapid is a prospective warning, not just a record of past loss.

Pulse	How it presents	Indicates	What to do about it
Flooding 洪 in winter	Wide, forceful, fades gently — out of season	Excess Yang building against the seasonal grain	Schrier's teaching case: untreated → cerebral hemorrhage the following summer.
Minute 微	Extremely thin AND forceless, fades in and out	Severe deficiency of Qi and Blood	One step beyond Fine. Treat as depletion, not simple Blood Def.
Knotted + Wiry	Slow, irregular stops, taut	Disease of the arteries, or hypertension	Named explicitly on the Knotted combinations slide.

The governing principle

Rhythm irregularities and shape-loss findings (Intermittent, Scattered, Hasty-Fine-weak) are the ones that most often mean something structural. A pulse that you cannot count, cannot find, or that stops on a schedule is a referral conversation, not a treatment-principle conversation.

PART VI · Pulse Blocks — Ying Cycle

What a pulse block is

Qi moving through the twelve channels can hold up at a junction. The pulse reads it the way a pinched garden hose reads: **full and tight before the pinch, weak and relatively empty after it**. Treatment is to needle the **exit point** of the backed-up channel and the **entry point** of the empty one, reopening the passage.

Why this requires writing pulses down

You cannot see a block by feeling one position at a time and never recording it. A block is a **relationship between adjacent positions** — it only appears when you can compare. This is Schrier's stated reason for keeping a written pulse record.

1 The Ying Cycle — Order of Flow

The sequence, which is also the organ clock

LU → LI → ST → SP → HT → SI → BL → KD → PC → TB → GB → LR → back to LU

Twelve channels, twelve months. The cycle begins with the **Lung** — first breath, and sunrise.

2 Mapping the Cycle Onto the Six Positions

Position	Organs held	Adjacent in the Ying cycle
L Cùn	HT · SI	HT → SI is a direct junction
L Guān	LR · GB	GB → LR is a direct junction (and LR → LU closes the cycle)
L Chǐ	KD · BL	BL → KD is a direct junction
R Cùn	LU · LI	LU → LI opens the cycle
R Guān	SP · ST	ST → SP is a direct junction
R Chǐ	PC · TB	PC → TB is a direct junction

What you're comparing

Walk the cycle in order and ask at each step: is the flow smooth, or does one side pile up while the next runs thin? You are not naming a quality here — you are comparing **relative fullness between neighbours**.

3 Block-Finding Worksheet

How to run it

Grade each position 0–5 for relative fullness (0 = empty, 5 = full and tight). Then read down the cycle order and mark any step where a high number is immediately followed by a low one. That step is your candidate block.

Reading 1 Date _____ Subject _____ Time _____

Cycle step	Fullness 0–5	Next step	Fullness 0–5	Drop?	Suspected block
LU		LI			
LI		ST			
ST		SP			
SP		HT			
HT		SI			
SI		BL			
BL		KD			
KD		PC			
PC		TB			
TB		GB			
GB		LR			
LR		LU			

Largest drop found (from → to):

Exit / entry points I would consider:

Signs & symptoms that corroborate (body / mind / spirit):

Reading 2 Date _____ Subject _____ Time _____

Cycle step	Fullness 0–5	Next step	Fullness 0–5	Drop?	Suspected block
LU		LI			
LI		ST			
ST		SP			
SP		HT			
HT		SI			
SI		BL			
BL		KD			
KD		PC			
PC		TB			
TB		GB			
GB		LR			
LR		LU			

Largest drop found (from → to):

Exit / entry points I would consider:

Signs & symptoms that corroborate (body / mind / spirit):

Reading 3 Date _____ Subject _____ Time _____

Cycle step	Fullness 0–5	Next step	Fullness 0–5	Drop?	Suspected block
LU		LI			
LI		ST			
ST		SP			
SP		HT			
HT		SI			
SI		BL			
BL		KD			
KD		PC			
PC		TB			
TB		GB			
GB		LR			
LR		LU			

Largest drop found (from → to):

Exit / entry points I would consider:

Signs & symptoms that corroborate (body / mind / spirit):

Reading 4 Date _____ Subject _____ Time _____

Cycle step	Fullness 0–5	Next step	Fullness 0–5	Drop?	Suspected block
LU		LI			
LI		ST			
ST		SP			
SP		HT			
HT		SI			
SI		BL			
BL		KD			
KD		PC			
PC		TB			
TB		GB			
GB		LR			
LR		LU			

Largest drop found (from → to):

Exit / entry points I would consider:

Signs & symptoms that corroborate (body / mind / spirit):

4 Husband–Wife Block

Know the term, hold the rule loosely

Worsley's definition: the **left pulse weaker in quality and quantity than the right**. Schrier flags the naming as dated and notes the classical texts say the reverse for women — right stronger, right being Yin, left Yang. He is openly skeptical the rule holds universally. Recognise it if someone uses the term; don't build a diagnosis on it alone.

Check	L total	R total	Which side stronger	Notes

PART VII · Bridge to Week 4 — Tongue

Purpose

Week 4 moves to tongue diagnosis, but pulse doesn't stop — the two are read together, and Schrier already seeded tongue material inside the pulse lectures. This bridges what you just learned into what's coming, so Week 4 costs less.

1 What Already Carried Over

Taught during pulse weeks	Belongs to tongue
Confirming a Rapid pulse by looking for a red tongue and yellow coating	Coating colour → Heat
Confirming Slippery by checking for a sticky/greasy coating	Coating texture → Damp/Phlegm
Coffee yellowing a coating — "does it mean Heat? Maybe, maybe not"	Confounders and inquiry
Deviated tongue = Internal Wind (check trauma first)	Tongue body shape
Medications altering the tongue (all 7 categories)	Coating and body confounders
The tongue map zones (tip/sides/centre/root)	Zone-to-organ correspondence

2 The Structural Parallel

Learn tongue as the same shape of system you just learned

Pulse decomposed into **location / speed / strength** and then combined. Tongue decomposes the same way — and the combinations work identically.

Pulse axis	Tongue equivalent	What both are telling you
Floating / Deep — location	Coating 苔 vs. Body 體 — surface vs. substrate	Coating reflects pathogens and Yang organs; body reflects the true underlying condition
Rapid / Slow — speed = Heat/Cold	Red / Pale — colour = Heat/Cold	Same axis, different sense
Full / Empty — strength	Thick / Thin coating; swollen vs. thin body	Excess vs. Deficiency
Slippery — Damp/Phlegm	Sticky, greasy coating	Same pathogen, two pillars
Choppy / Fine — Blood Def	Pale body, thin body	Same pattern, two pillars

3 Where Pulse and Tongue Disagree

This is the highest-value skill in the Four Pillars

When pulse and tongue agree, your diagnosis is confirmed and nothing interesting happened. When they **disagree**, that's the diagnostic information — it's how you catch False Heat and False Cold.

Pulse says	Tongue says	Likely reading
Floating-Empty, cold limbs	Pale and wet	True Cold / False Heat — the flush is superficial
Deep and Forceful, cold limbs	Red with yellow coat	True Heat / False Cold — Heat drives Yang to the extremities
Rapid	Pale, no Heat signs	Check confounders — recent exertion? Anxiety? Then re-take.
Slippery	No coating change, no heaviness	Food retention? Pregnancy? Ask before concluding Phlegm.
Normal / Huǎn	Markedly abnormal	Trust the tongue for chronicity — the tongue changes slowly, the pulse fast

The general rule on timescale

The pulse changes quickly; the tongue changes slowly. A pulse can shift within a single rotation of students. A tongue body reflects a longer-standing state. When they conflict on chronicity, that difference is usually the explanation.

4 Prime Yourself Before Week 4

Three things to do in the week before tongue starts

1. Keep journaling — every entry already has both tongue and pulse fields, so you'll arrive with data rather than a blank page.
2. Look at your own tongue daily in good natural light. Note body colour, coating, and any change. You need a baseline you know well.
3. Re-read the medication → tongue material — it was taught in Week 2 and will be assumed knowledge.

One practical warning Schrier raised

Photographs of tongues are unreliable — screens, printing, and glossy paper all shift the colour, and the same tongue looks different projected, on your laptop, and in the room. Prioritise live observation over images wherever you can.

5 Carry-Forward Checklist

- ☐ I can name all 6 pulse positions (Nan Jing) without looking
- ☐ I can find superficial, middle, and deep at each position
- ☐ I can identify Floating, Deep, Slow, Rapid, Full by hand
- ☐ I know which 3 findings are referral-level
- ☐ I have at least 10 journal entries with both tongue and pulse recorded
- ☐ I know the 4 tongue zones and their organs
- ☐ I know what the coating vs. the body each reflect